

Background:

- text-heavy. Slide would be more impactful with a figure/graph
 - e.g. screenshot of research paper used to discuss the background
- EDSS scale: no need to mention, as it's not discussed for the result in the presentation
- Can also talk about where we get vitamin D

Knowledge gap:

- Add slides with research question and hypothesis
- Could have schematic/ graphic for patient with MS, etc

Statistical test slide: remove

- What did you learn about the biology of the disease, mention key things from the statistical test while talking about each result
- How the aim designed to answer the research question
- Not going into too much about the thought process for the aim

Aim slides: no need to include, a bit repetitive when mentioning it in the result again

Intention of analysis → result of analysis (intention can be verbally described)

Alpha diversity 1:

- (for all result slides) more descriptive title
- (title example) vit d supplementation is significantly associated with shannon diversity
- Help audience to understand a lot faster & have guidance
- If some stuff is described in the modules, no need to explain again
- E.g. we can keep shannon diversity explanation, as it gives the comparison with the following slide. (no need to include the definition of alpha diversity)

Alpha diversity 2:

- no need to include the numerical p-value, just say whether it's significant or not

Beta diversity:

- Text is a bit small, spacing things out a bit more, increase text size
- Have a bullet point saying there is no significance identified

Indicator species:

- Combine the indicator species slides together for comparison/remove the significant taxa slide (the bar plot one), make it a bullet point to explain is good.
- Scale of y-axis need to adjust
- Remove the title for the box plot
- The indicator value is low, which is why the box would stack at the bottom
- Indicator species: the succinivibrio there might not be a taxa, might phrase it as “it’s the best result we can find”, indicator value of 0.17 might not be good enough for the indicator taxa
- No need to rarefy. Even though it’s not a indicator taxa, there is difference in the in between supplementation and no supplementation

Consistent for axis label for vitamin D supplementation:

- yes/no or supplement/no supplement

Change the color as well

Differential abundance:

- Fring: scfa producer
- Ery: associate with fat metabolism
- Animate for each bulletpoint
- Title to summarize the result
- Desulfovibrio: significant interest in the field right now, also look into it with parkinson disease. (neuro related)
 - Mention the PD in the manuscript as well
- Arrow with → enrich; ←depleted
- Straightforward description for each taxa

Functional analysis:

- Lfc2 of 0.5 is fine
- Cite the pathway

Correlation matrix 1:

- Keep it in supplement for the manuscript
- Move this into previous part of the presentation
- No need to stick to the aim, better to mention it earlier on
- Cut off for significance: 0.2-0.3 for getting significant p-value
- Can say “These one are more associated with each other” for those that don’t meet the cut off

Correlation matrix 2:

- Keep this one for discussion for aim 4
- Explain the heat map (e.g. green = more positive correlated, red=more negative correlated etc)

Conclusion:

- Current points work
- First point is slightly incorrect, as beta shows no significant difference
- Talk about why we got certain result. What is expected and not expected
- No need to mention treatment_status.
- Could include limitation in the conclusion slide

Future direction (approx 3 points):

- #3 is a bit vague.
- Potential wet lab work/dry lab work: e.g. experiment/ human test
- 2 is good, pointing supplementation

Overall:

- Slide: citing information (eg. ASM citation)
- Reference slide format(first author, publication)
- Animation could be done when describing each bulletpoint